

數字經濟核心科技系列：前端網站開發人員課程

## (一) 初階網絡程式設計

### 2. 超文字標示語言(HTML)

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# 4. HTML：高階

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## 4. HTML高階

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## 4. HTML高階

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# 4.1 Revise

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## 4. HTML高階

# 4.1 Revise

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- 表格的標籤和屬性是什麼？
- 如何創建一個工具提示？
- 你能說出任何特殊效果嗎？
  - 它們的標籤和屬性是什麼？

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# 4.2 什麼是Bootstrap ( BS ) ?

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#### 4. HTML高階

## 4.2 什麼是Bootstrap ( BS ) ?

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- 最新版本是BS5
- 它是最流行的免費CSS框架，用於開發響應式和移動優先的網站
- 作為參考，<https://www.w3schools.com/bootstrap4/default.asp>
- <https://getbootstrap.com/2.3.2/components.html#dropdowns>

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# 4.3 Bootstrap elements

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## 4. HTML高階

# 4.3 Bootstrap elements

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- Grid 網格
- Alerts 警示
- Spinner輪轉器
- Table 表格

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#### 4. HTML高階

### 4.3.1 Grid網格

- 在 Bootstrap 4 (BS4) 中，網格系統 (Grid System) 採用 12 欄的設計

|           |           |           |           |           |           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 | span<br>1 |
| span 4    |           |           |           | span 4    |           |           |           | span 4    |           |           |           |
| span 4    |           |           |           | span 8    |           |           |           |           |           |           |           |
| span 6    |           |           |           |           |           | span 6    |           |           |           |           |           |
| span 12   |           |           |           |           |           |           |           |           |           |           |           |

## 4. HTML高階

# 4.3.1 網格

- BS4的網格系統有5個等級:
  1. .col- (超小設備--屏幕寬度小於576px)
  2. .col-sm- (小型設備-屏幕寬度等於或大於576px)
  3. .col-md- (中型設備-屏幕寬度等於或大於768px)
  4. .col-lg- (大型設備-屏幕寬度等於或大於992px)
  5. .col-xl- (超大型設備--屏幕寬度等於或大於1200px)

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## 4. HTML高階

### 4.3.1 網格

- 要在 <head> 中引用 Bootstrap 4 (BS4) 的 CSS 和 JS，你可以直接使用官方 CDN

```
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
```

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## 4. HTML高階

# 4.3.1 網格

- 調整瀏覽器的大小，看看效果如何
- 當屏幕寬度小於576px時，各欄會自動相互堆疊

```
<!-- Grid -->
<h3>Grid</h3>
<h4>Responsive Columns</h4>
  <div class="row">
    <div class="col-sm-3">.col-sm-3</div>
    <div class="col-sm-3">.col-sm-3</div>
    <div class="col-sm-3">.col-sm-3</div>
    <div class="col-sm-3">.col-sm-3</div>
  </div>
```

.col-sm-3  
.col-sm-3  
.col-sm-3  
.col-sm-3

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## 4. HTML高階

### 4.3.1 網格

- 調整瀏覽器的大小，看看效果如何
- 它們是響應式的，但不會相互堆疊

```
<h4>Equal width columns</h4>
<div class="row">
  <div class="col">.col</div>
  <div class="col">.col</div>
  <div class="col">.col</div>
</div>
```

col

.col

.col

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## 4. HTML高階

# 4.3.1 網格：自己嘗試一下

- 嘗試添加更多的列!

```
<div class="col">.col</div>
```

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#### 4. HTML高階

### 4.3.1 網格

- 你可以透過在 class 名稱後加上數字來指定欄位寬度
- 每一行 (row) 的總和必須  $\leq 12$ ，否則排版會出現問題

```
<h4>Unequal width columns</h4>
<div class="row">
  <div class="col-6">.col-6</div>
  <div class="col-6">.col-6</div>
</div>
```

.col-6

.col-6

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#### 4. HTML高階

### 4.3.1 網格

```
<div class="row">
  <div class="col-1">.col-1</div>
  <div class="col-11">.col-11</div>
</div>
```

.col-1    .col-11

- 由於他們只用了5+1列，右邊的6列將是空的。

```
<div class="row">
  <div class="col-5">.col-5</div>
  <div class="col-1" style="background-color: grey;">.col-1</div>
</div>
```

.col-5

.col-1

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4. HTML高階

## 4.3.1 網格：自己嘗試一下

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- 嘗試用不同的列跨度添加更多的行!

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## 4. HTML高階

# 4.3.2 Alerts 警示

### • 8種類型的alerts 警示:

```
<!-- Alerts -->
<h3>Alerts</h3>
<p>Alerts are created with the .alert class, followed by a contextual color classes:</p>
<div class="alert alert-success">
  <strong>Success!</strong> This alert box could indicate a successful or positive action.
</div>
<div class="alert alert-info">
  <strong>Info!</strong> This alert box could indicate a neutral informative change or action.
</div>
<div class="alert alert-warning">
  <strong>Warning!</strong> This alert box could indicate a warning that might need attention.
</div>
<div class="alert alert-danger">
  <strong>Danger!</strong> This alert box could indicate a dangerous or potentially negative action.
</div>
<div class="alert alert-primary">
  <strong>Primary!</strong> Indicates an important action.
</div>
<div class="alert alert-secondary">
  <strong>Secondary!</strong> Indicates a slightly less important action.
</div>
<div class="alert alert-dark">
  <strong>Dark!</strong> Dark grey alert.
</div>
<div class="alert alert-light">
  <strong>Light!</strong> Light grey alert.
</div>
```

## Alerts

Alerts are created with the .alert class, followed by a contextual color classes:

**Success!** This alert box could indicate a successful or positive action.

**Info!** This alert box could indicate a neutral informative change or action.

**Warning!** This alert box could indicate a warning that might need attention.

**Danger!** This alert box could indicate a dangerous or potentially negative action.

**Primary!** Indicates an important action.

**Secondary!** Indicates a slightly less important action.

**Dark!** Dark grey alert.

**Light!** Light grey alert.

## 4. HTML高階

### 4.3.2 Alerts : .alert-link

- 加入 `<a class="alert-link">` 以建立連結
- 你可以在 href 屬性中放置任何超連結或其他 HTML 檔案

```
<div class="alert alert-success">  
  <strong>Success!</strong> Click <a href="#" class="alert-link">here</a> to read.  
</div>
```

**Success!** Click **here** to read.

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#### 4. HTML高階

### 4.3.2 Alerts : Close button關閉按鈕

- 帶有 class="close" 以及 data-dismiss="alert" 的按鈕用來關閉警示框
- alert-dismissible 類別會為關閉按鈕增加額外的內距 (padding)

```
<div class="alert alert-success alert-dismissible">  
  <button type="button" class="close" data-dismiss="alert">&times;</button>  
  <strong>Success!</strong> Click <a href="#" class="alert-link">here</a> to read.  
</div>
```

**Success!** Click **here** to read.



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## 4. HTML高階

# 4.3.2 Alerts : Fade out淡出

- fade 與 show 類別在關閉警示框時會加入淡出效果

```
<div class="alert alert-success alert-dismissible fade show">  
  <button type="button" class="close" data-dismiss="alert">&times;</button>  
  <strong>Success!</strong> Click <a href="#" class="alert-link">here</a> to read.  
</div>
```

**Success!** Click **here** to read.



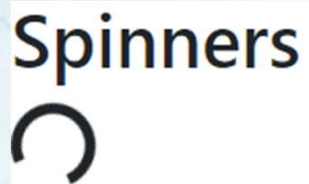
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#### 4. HTML高階

### 4.3.3 Spinners輪轉器

- 也叫載入指示器
- 可以透過加入 .spinner-border 來建立

```
<!-- Spinners -->  
<h3>Spinners</h3>  
<div class="spinner-border"></div>
```



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## 4. HTML高階

# 4.3.3 Spinners輪轉器

- 其他9種類型的輪轉器:

```
<div class="spinner-border text-muted"></div>  
<div class="spinner-border text-primary"></div>  
<div class="spinner-border text-success"></div>  
<div class="spinner-border text-info"></div>  
<div class="spinner-border text-warning"></div>  
<div class="spinner-border text-danger"></div>  
<div class="spinner-border text-secondary"></div>  
<div class="spinner-border text-dark"></div>  
<div class="spinner-border text-light"></div>
```



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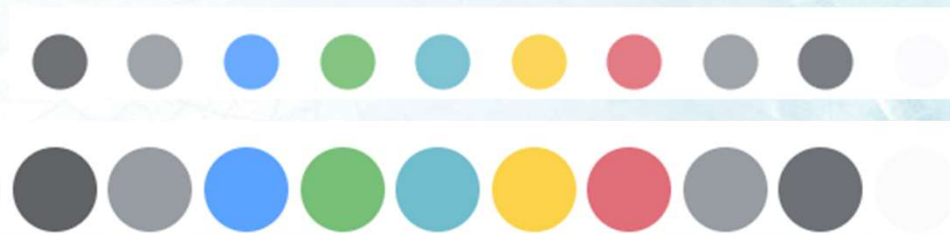


#### 4. HTML高階

### 4.3.3 Spinners : Growing成長型

- .spinner-grow
- 以放大 (grow) 取代旋轉 (spin)

```
<div class="spinner-grow"></div>  
<div class="spinner-grow text-muted"></div>  
<div class="spinner-grow text-primary"></div>  
<div class="spinner-grow text-success"></div>  
<div class="spinner-grow text-info"></div>  
<div class="spinner-grow text-warning"></div>  
<div class="spinner-grow text-danger"></div>  
<div class="spinner-grow text-secondary"></div>  
<div class="spinner-grow text-dark"></div>  
<div class="spinner-grow text-light"></div>
```



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## 4. HTML高階

### 4.3.3 Spinners: Small

- 加入.spinner-border-sm / .spinner-grow-sm以建立小型輪轉器

```
<div class="spinner-border spinner-border-sm"></div>  
<div class="spinner-grow spinner-grow-sm"></div>
```



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#### 4. HTML高階

### 4.3.3 Spinners : Buttons按鈕

- 將輪轉器 (Spinner) 放置在按鈕中

```
<button class="btn btn-primary">  
  <span class="spinner-border spinner-border-sm"></span>  
</button>
```



- 將輪轉器 (Spinner) 放置在停用的按鈕中

```
<button class="btn btn-primary" disabled>  
  <span class="spinner-grow spinner-grow-sm"></span>  
  Loading..  
</button>
```



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## 4. HTML高階

# 4.3.4 Table表格

- 我們已經學習了<table>，但BS4對它有更好的設計!

```
<!-- Table -->
<h3>Table</h3>
<table class="table">
  <thead>
    <tr>
      <th>Firstname</th>
      <th>Lastname</th>
      <th>Email</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Saint</td>
      <td>Mary</td>
      <td>saint@email.com</td>
    </tr>
    <tr>
      <td>John</td>
      <td>Goofy</td>
      <td>john@email.com</td>
    </tr>
    <tr>
      <td>April</td>
      <td>Dairy</td>
      <td>april@email.com</td>
    </tr>
  </tbody>
</table>
```

Table

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Saint     | Mary     | saint@email.com |
| John      | Goofy    | john@email.com  |
| April     | Dairy    | april@email.com |

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#### 4. HTML高階

### 4.3.4 Table : Stripe條紋

- 加入 .table-striped 以讓表格呈現斑馬條紋效果

```
<table class="table table-striped">
```

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Saint     | Mary     | saint@email.com |
| John      | Goofy    | john@email.com  |
| April     | Dairy    | april@email.com |

#### 4. HTML高階

### 4.3.4 Table : Border邊框

- 加入 .table-bordered 以在每個儲存格加上邊框

```
<table class="table table-striped table-bordered">
```

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Saint     | Mary     | saint@email.com |
| John      | Goofy    | john@email.com  |
| April     | Dairy    | april@email.com |

#### 4. HTML高階

### 4.3.4 Table : Borderless無邊框

- 將

```
<table class="table table-striped table-borderless">
```

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Saint     | Mary     | saint@email.com |
| John      | Goofy    | john@email.com  |
| April     | Dairy    | april@email.com |



#### 4. HTML高階

### 4.3.4 Table : Hover effect懸停效果

- 加入 .table-hover 以啟用互動式滑鼠懸停效果

```
<table class="table table-striped table-bordered table-hover">
```

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Saint     | Mary     | saint@email.com |
| John      | Goofy    | john@email.com  |
| April     | Dairy    | april@email.com |

#### 4. HTML高階

### 4.3.4 Table

- 加入.table-dark，使該表變暗

```
<table class="table table-striped table-bordered table-hover table-dark">
```

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Saint     | Mary     | saint@email.com |
| John      | Goofy    | john@email.com  |
| April     | Dairy    | april@email.com |

## 4. HTML高階

# 4.3.4 Table：Contextual具語境的

- 共有10種類型的表格（如alerts/spinners）：

```
</table>
<table class="table">
  <thead>
    <tr>
      <th>Firstname</th>
      <th>Lastname</th>
      <th>Email</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Default</td>
      <td>Lastname</td>
      <td>def@email.com</td>
    </tr>
    <tr class="table-primary">
      <td>Primary</td>
      <td>Lastname</td>
      <td>joe@email.com</td>
    </tr>
    <tr class="table-success">
      <td>Success</td>
      <td>Lastname</td>
      <td>john@email.com</td>
    </tr>
    <tr class="table-danger">
      <td>Danger</td>
      <td>Lastname</td>
      <td>mary@email.com</td>
    </tr>
    <tr class="table-info">
      <td>Info</td>
      <td>Lastname</td>
      <td>july@email.com</td>
    </tr>
    <tr class="table-warning">
      <td>Warning</td>
      <td>Lastname</td>
      <td>bo@email.com</td>
    </tr>
    <tr class="table-active">
      <td>Active</td>
      <td>Lastname</td>
      <td>act@email.com</td>
    </tr>
    <tr class="table-secondary">
      <td>Secondary</td>
      <td>Lastname</td>
      <td>sec@email.com</td>
    </tr>
    <tr class="table-light">
      <td>Light</td>
      <td>Lastname</td>
      <td>angie@email.com</td>
    </tr>
    <tr class="table-dark text-dark">
      <td>Dark</td>
      <td>Lastname</td>
      <td>bo@example.com</td>
    </tr>
  </tbody>
</table>
```

printing will not be tolerated.

## 4. HTML高階

# 4.3.4 Table : Contextual

- 共有10種類型的表格（如alerts/spinners）

| Firstname | Lastname | Email           |
|-----------|----------|-----------------|
| Default   | Lastname | def@email.com   |
| Primary   | Lastname | joe@email.com   |
| Success   | Lastname | john@email.com  |
| Danger    | Lastname | mary@email.com  |
| Info      | Lastname | july@email.com  |
| Warning   | Lastname | bo@email.com    |
| Active    | Lastname | act@email.com   |
| Secondary | Lastname | sec@email.com   |
| Light     | Lastname | angie@email.com |
| Dark      | Lastname | bo@example.com  |

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# 4.4 練習

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## 4. HTML高階

# 4.4 練習

## 4. HTML: Advanced

Navbar [Google](#) [Yahoo](#)

**Notice!** Click [here](#) to read.



Don't use style, use grid to put me in the center

New In

| ID  | Items               | Price |     |
|-----|---------------------|-------|-----|
| 001 | Double Layer Tee    | £     | 239 |
| 002 | Blocking Pocket Tee | £     | 258 |
| 003 | Graphic Print Tee   | £     | 258 |

- Grid
- Span 5

- Grid
- Span 4

- Grid
- Span 3

[Back to Top](#)

e tolerated.

#### 4. HTML高階

## 4.4 練習：建議的答案

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Exercise4</title>
5      <meta charset="UTF-8">
6      <meta http-equiv="refresh" content="30">
7      <meta name="viewport" content="width=device-width, initial-scale=1.0">
8      <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
9      <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
10     <script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
11     <script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
12 </head>
```

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## 4. HTML高階

# 4.4 練習：建議的答案

```
20      <!-- Nav -->
21      <nav>
22          <span class="col">Navbar</span>
23          <span class="col">
24              <a href="https://www.google.com/" alt="Google">Google</a>
25          </span>
26          <span class="col">
27              <a href="https://hk.yahoo.com/" alt="Yahoo">Yahoo</a>
28          </span>
29      </nav>
```

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## 4. HTML高階

# 4.4 練習：建議的答案

---

```
30      <!-- Alert -->
31  ✓    <div class="alert alert-warning alert-dismissible fade show">
32      <button type="button" class="close" data-dismiss="alert">&times;</button>
33      <strong>Notice!</strong> Click <a href="#" class="alert-link">here</a> to read.
34    </div>
```

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## 4. HTML高階

# 4.4 練習：建議的答案

```
35      <!-- Spinner -->
36      <div class="row">
37          <div class="col-6"></div>
38          <div class="spinner-border text-primary">0</div>
39          <p>Don't use style, use grid to put me in the center</p>
40          <div class="col-6"></div>
41      </div>
```

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## 4.4 練習：建議的答案

```
48 <table class="table table-dark table-hover table-bordered">
49   <thead>
50     <tr>
51       <th>ID</th>
52       <th>Items</th>
53       <th colspan="2">Price</th>
54     </tr>
55   </thead>
56   <tbody>
57     <tr>
58       <td>001</td>
59       <td>Double Layer Tee</td>
60       <td>&#8356;</td>
61       <td>239</td>
62     </tr>
63     <tr>
64       <td>002</td>
65       <td>Blocking Pocket Tee</td>
66       <td>&#8356;</td>
67       <td>258</td>
68     </tr>
```

nting will not be tolerated.

## 4. HTML高階

# 4.4 練習：建議的答案

```
69         <tr>
70             <td>003</td>
71             <td>Graphic Print Tee</td>
72             <td>&#8356;</td>
73             <td>258</td>
74         </tr>
75     </tbody>
76 </table>
77 <div class="row">
78     <div class="col-5">
79         <ul>
80             <li>Grid</li>
81             <li>Span 5</li>
82         </ul>
83     </div>
```

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## 4. HTML高階

# 4.4 練習：建議的答案

```
84      <div class="col-4">
85          <ul>
86              <li>Grid</li>
87              <li>Span 4</li>
88          </ul>
89      </div>
90      <div class="col-3">
91          <ul>
92              <li>Grid</li>
93              <li>Span 3</li>
94          </ul>
95      </div>
96  </div>
97  </main>
```

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## 4. HTML高階

# 4.4 練習：建議的答案

---

```
100      <!-- copyright -->
101      <div style="text-align: center;">@ 2021 Copyright: ABC Institute    </div>
102    </footer>
103  </body>
104  </html>
```

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#### 4. HTML高階

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<<https://www.w3schools.com/bootstrap4/default.asp>> [2026 年 3 月 12 日]
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